

Executive Summary

PAUHelper is a specialized Retrieval-Augmented Generation (RAG) web application designed to guide students through the complex, heavily regulated landscape of university entrance exams in Spain. Beyond its immediate educational utility, the project serves as a comprehensive demonstration of enterprise-grade AI deployment, showcasing how to securely bridge the gap between powerful Large Language Models (LLMs) and strictly validated proprietary data.

The Business Problem: High-Stakes Information Validation

Students preparing for the PAU (Prueba de Acceso a la Universidad) face a highly fragmented data environment. The curriculum, grading criteria, and exam formats are strictly regulated but vary dramatically depending on the student's autonomous community (region) and specific subject.

The Frictions & Risks:

- **Information Overload:** Official guidelines are buried in hundreds of pages of PDF bulletins scattered across various regional government websites.
- **AI Hallucination:** Generic AI tools (like standard ChatGPT) often provide "average" or generalized answers that do not align with precise regional criteria, risking the student's exam score.
- **Lack of Traceability:** When an AI provides a study guideline, the user currently has no way to verify which official mandate that guideline came from.

The Proposed Solution: Context-Aware Intelligence

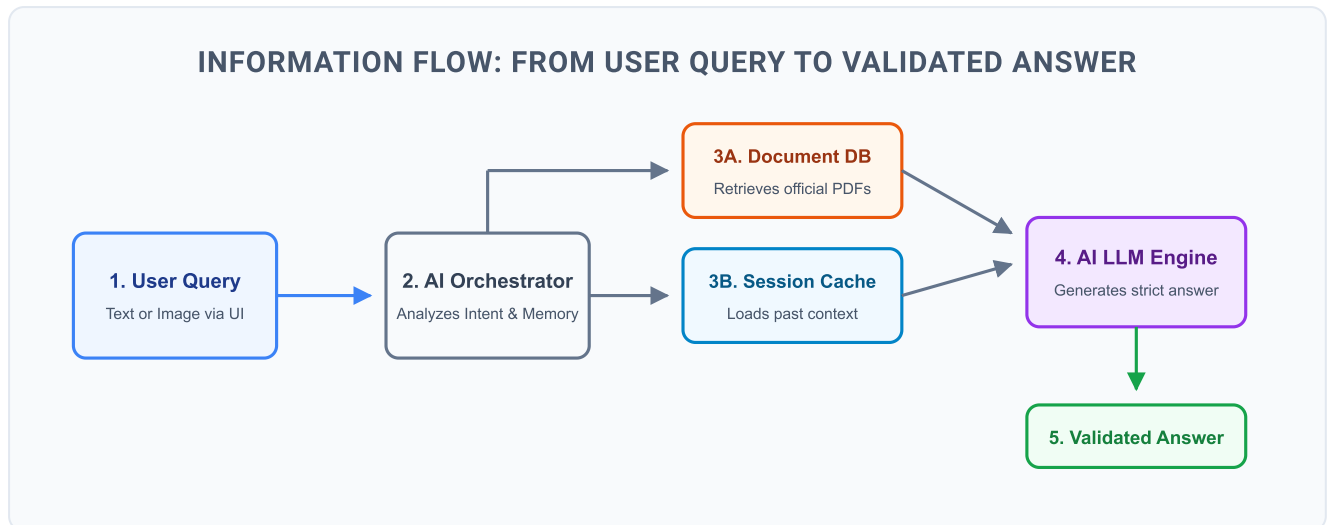
PAUHelper resolves these issues by acting as a hyper-specialized, highly reliable study assistant. Instead of relying on an AI's generalized parametric knowledge, the system forces the AI to read only verified, official documents before answering.

The User Experience:

- The student selects their specific region, subject, and language preferences.
- They ask a question or upload an image of a problem they are struggling with.
- The system instantly retrieves the exact official syllabus paragraphs matching the query, reads them, and returns an accurate answer strictly grounded in the official curriculum.
- Crucially, it provides the exact source document citations, establishing absolute trust.

System Architecture & Data Flow

To ensure zero hallucinations, PAUHelper separates the search for information from the generation of text. This orchestration is invisible to the non-technical user but guarantees data integrity.



1 User Request & Input Handling

The student selects their study configuration (Region, Subject, Language) and inputs a text query or uploads an image. The frontend securely packages this and sends it to the backend over an encrypted connection.

2 AI Orchestrator (LangGraph)

The backend acts as a smart traffic controller. It processes the query, identifies exactly what the student is looking for, and prepares a search strategy tailored specifically to the selected region and subject.

3 Data & Memory Retrieval

The system simultaneously queries a specialized Vector Database to pull the most relevant official document paragraphs, and checks a Redis Cache to recall the user's previous questions, ensuring a continuous, natural conversation.

4 Language Model Generation

The extracted documents, conversation history, and user question are bundled together and sent to the LLM (like Google Gemini). The model is strictly instructed: *"You may only answer using the documents provided."*

Cross-Industry Business Value

While PAUHelper is optimized for the education sector, the underlying architecture—Retrieval-Augmented Generation—solves one of the most critical challenges in the modern enterprise: **making vast amounts of unstructured proprietary data instantly queryable without compromising accuracy.**

Eliminating Information Silos

Knowledge workers spend roughly 20% of their day searching for internal information. By centralizing data into a semantic vector database, employees can query in natural language and receive synthesized answers instantly, drastically reducing research time and operational friction.

Mitigating AI Hallucinations

Standard LLMs are trained to generate plausible-sounding text, which can lead to costly errors in business environments. Because this architecture is restricted to utilizing only provided internal documents, the risk of "invented" facts is virtually eliminated, enabling enterprise trust.

Parallel Application: Legal Tech

Just as a student queries regional exam rules, lawyers can use this exact architecture to query case law, contract portfolios, and compliance guidelines. The system retrieves relevant clauses and explicitly cites the specific contract page, accelerating due diligence.

Parallel Application: Customer Support

Corporate support teams can deploy this as an "Agent Copilot." Instead of educational syllabi, the database is loaded with internal technical manuals and past support tickets, allowing tier-1 agents to instantly resolve complex client issues with validated, accurate data.

The Data Flywheel Effect

As users interact with the system, the application inherently tracks which queries frequently fail to match existing documents. This provides business leadership with automated insights into "knowledge gaps" within their corporate documentation, driving targeted updates to company manuals, policies, or client-facing FAQs.

Technology Stack & Scalability

To achieve enterprise-grade reliability, PAUHelper is built on a modern, cloud-native technology stack. It is designed to be highly scalable, cost-efficient, and easily maintainable by a small engineering team.

Frontend: React & Vite

Delivers a seamless, Single Page Application (SPA) experience directly in the browser. It handles complex state (like chat history and dynamic image uploads) smoothly, ensuring users experience zero friction, regardless of the device they use.

Backend: FastAPI & LangGraph

FastAPI provides a high-performance, async API gateway. LangGraph handles the complex state machine of the AI agent, explicitly routing tasks (retrieve, generate, summarize memory) to ensure the AI follows a rigid, predictable logic flow.

Database: Qdrant Vector DB

Traditional databases search by keyword matching. Qdrant searches by mathematical "meaning." If a user asks about "writing rules," it will find documents talking about "syntax guidelines," ensuring high-quality retrieval even with varied phrasing.

Infrastructure: Google Cloud & Terraform

The entire environment is codified using Terraform (Infrastructure as Code). This ensures the infrastructure is perfectly reproducible, self-documenting, and auditable, completely eliminating manual server configuration errors.

Security & Cost-Efficiency (Serverless Scale-to-Zero)

Strict Network Isolation: The core database containing proprietary documents is deployed in a private Virtual Private Cloud (VPC) with no public IP addresses. It can only be accessed internally by authorized microservices.

Elastic Compute: Hosted on Google Cloud Run, the application utilizes a serverless architecture. During quiet hours (e.g., 3:00 AM), the application scales down to zero, costing nothing. During the weeks immediately preceding final exams, the service can instantly spin up hundreds of container clones to handle thousands of concurrent students without degradation in speed.